

ucsd_stereo_21deg_12d

Processed on: 2026-04-15 14:54:41

ASP version: 3.7.0-alpha

asp_plot version: 3.1.0

DEM Summary

Property	Value
DEM File	run-DEM.tif
Dimensions (px)	2499 x 2491
GSD (m)	1.20
CRS	EPSG:32611
Nodata (%)	14.4
Elevation Range (m)	-75.9 to 132.0
Acquisition Dates	2015-02-12 18:39:26 UTC; 2015-02-24 18:31:34 UTC

Processing Parameters

Runtime Summary

Step	Runtime
Bundle Adjust	0 hours and 16 minutes
Stereo	0 hours and 28 minutes
point2dem	0 hours and 0 minutes

Reference DEM:

ref/cop30_ucsd_wgs84_utm.tif

Bundle Adjust Command:

```
bundle_adjust --threads 8 --ip-per-image 10000 --tri-weight 0.1 --tri-robust-threshold 0.1 --camera-weight 0
1040010007A93700_P001.NTF 1040010007CA4D00_P001.NTF 1040010007A93700_P001.xml 1040010007CA4D00_P001.xml -o ba/run
```

Mapproject Commands:

Reconstructed from output GeoTIFF metadata (mapproject writes no log file); session type and grid size are the resolved values. Re-running reproduces the output grid exactly with ASP >= 3.7.0.

```
mapproject -t rpc --t_srs EPSG:32611 --tr 0.315 --t_projwin 475999.965 3635600.22 478999.71 3638600.28 --bundle-adjust-
prefix ba/run ref/cop30_ucsd_wgs84_utm.tif 1040010007A93700_P001.NTF 1040010007A93700_P001.xml
1040010007A93700_P001_map.tif
```

```
mapproject -t rpc --t_srs EPSG:32611 --tr 0.315 --t_projwin 475999.965 3635600.22 478999.71 3638600.28 --bundle-adjust-
prefix ba/run ref/cop30_ucsd_wgs84_utm.tif 1040010007CA4D00_P001.NTF 1040010007CA4D00_P001.xml
1040010007CA4D00_P001_map.tif
```

Stereo Command:

```
stereo --stereo-algorithm asp_mgm --subpixel-mode 9 --alignment-method none --bundle-adjust-prefix ba/run
1040010007A93700_P001_map.tif 1040010007CA4D00_P001_map.tif 1040010007A93700_P001.xml 1040010007CA4D00_P001.xml
stereo/run ref/cop30_ucsd_wgs84_utm.tif --corr-seed-mode 1 --sgm-collar-size 256 --compute-point-cloud-center-only
--threads 8
```

point2dem Command:

```
point2dem --tr 1.2 --t_srs EPSG:32611 --errorimage stereo/run-PC.tif
```

Report Generation Command:

```
asp_report --directory /Users/ben/Desktop/asp-plot-examples/ucsd_stereo_21deg_12d/ --bundle-adjust-prefix ba/ --stereo-
directory stereo/ --subset-km 0.25 --reuse-selections /Users/ben/Desktop/uw-github/asp_plot/reports/WorldView_UCSD-asp-
report_figure_selections.yml --report-filename /Users/ben/Desktop/uw-github/asp_plot/reports/WorldView_UCSD-asp-
report.pdf
```

Input Scenes

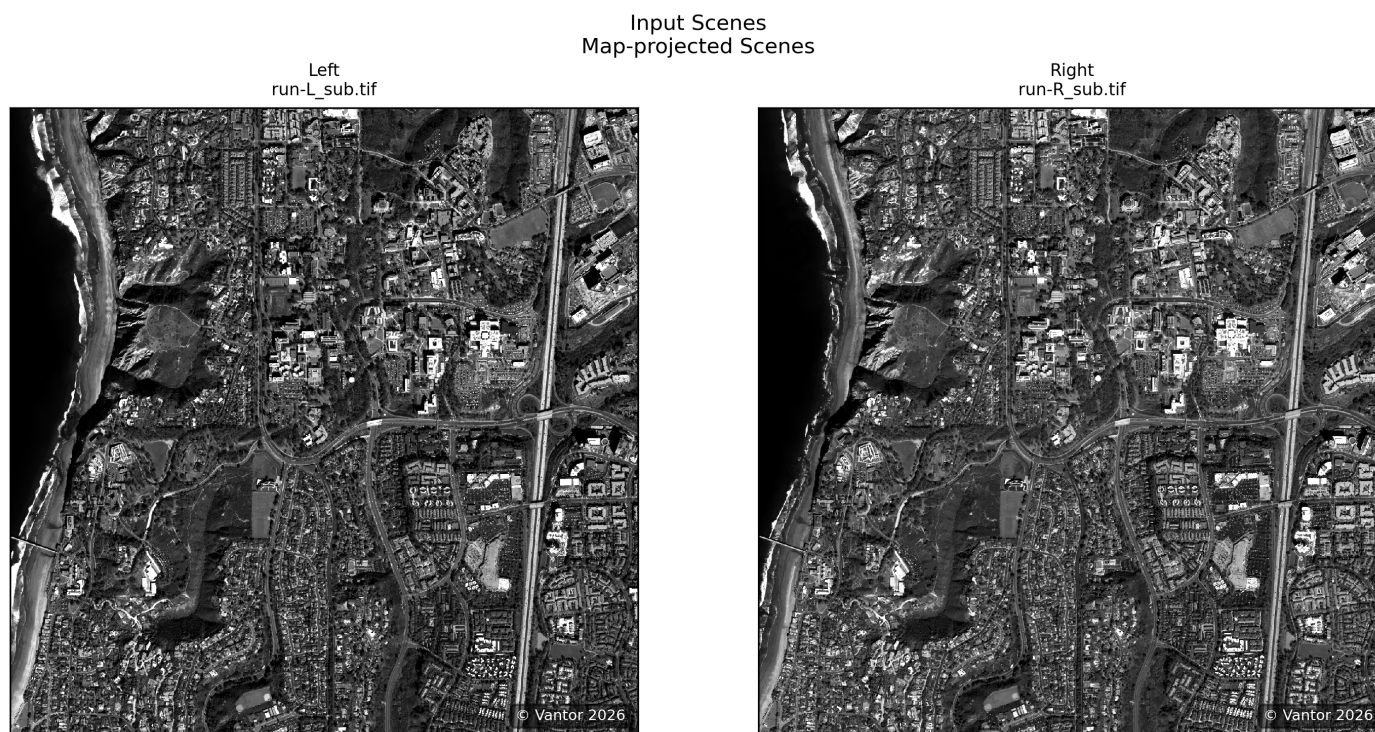


Figure 1: Left and right input scenes used for stereo processing. Non-mapprojected scenes are shown after ASP's alignment step (e.g., *affineepipolar*), which rotates images to create horizontal epipolar lines for correlation. Mapprojected scenes have been orthorectified with RPCs against a reference DEM to roughly align the two images prior to correlation, which reduces the disparity search range; they are displayed here in their map-projected orientation.

Stereo Geometry

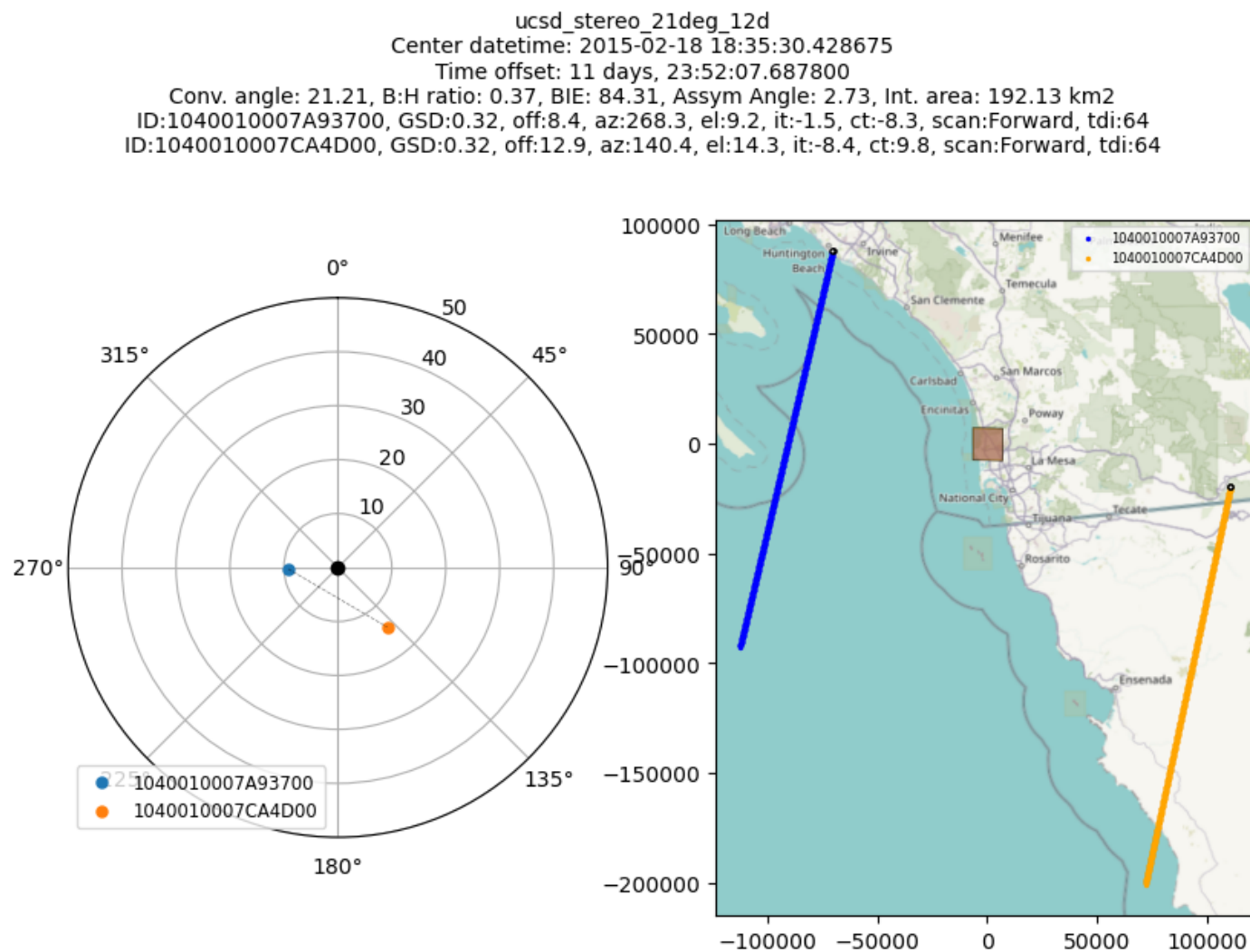


Figure 2: Stereo acquisition geometry skyplot and map view showing satellite viewing angles and scene footprints.

Match Points

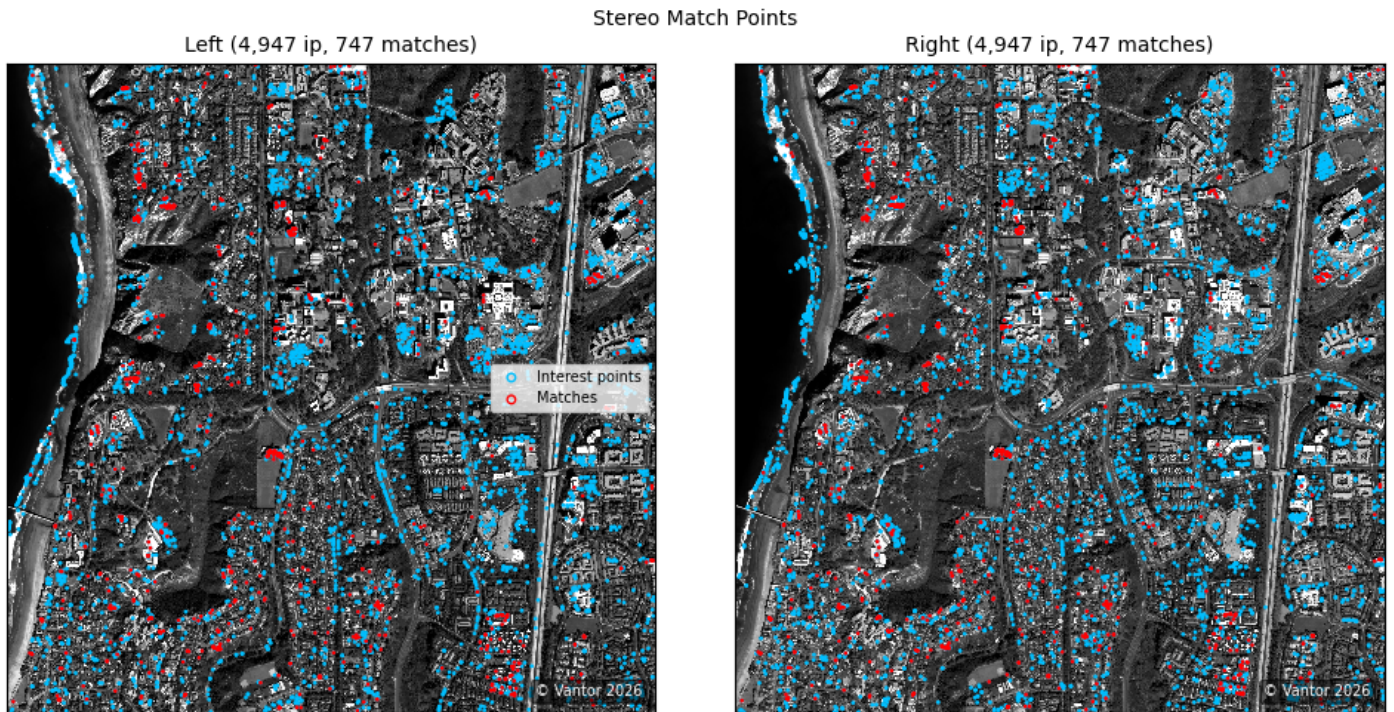


Figure 3: Interest point matches between left and right images (red), overlaid on the raw interest points detected on each image individually (blue, when the .vwip files are present in the stereo directory). These are produced by `stereo_corr` during its initial interest point matching step, which is used to set the search windows for subsequent dense correlation (not the dense correlation matches themselves). Comparing the raw interest points against the matches shows whether sparse matching traces back to poor matching or to areas with no detected interest points at all (e.g., limited texture, clouds, water). Some ASP configurations only write a .vwip file for the left image (the right image's interest points are detected on a temporary aligned image and not saved), in which case only matches are shown on the right.

Bundle Adjust Residuals (Log Scale)

Bundle Adjust Initial and Final Residuals (Log Scale)

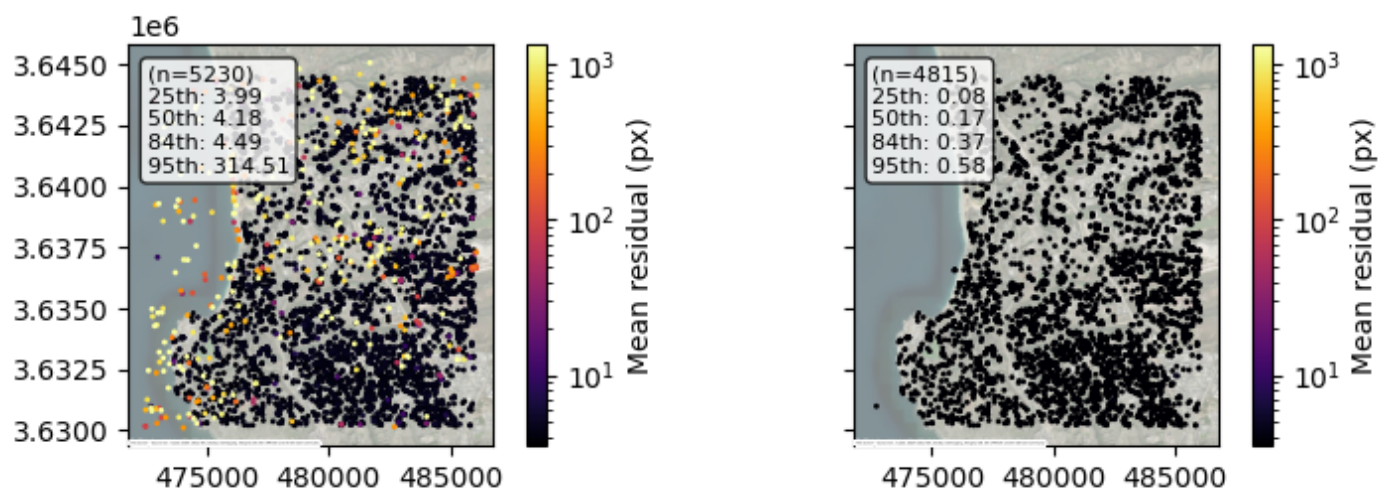


Figure 4: Initial and final bundle adjustment residuals on a logarithmic scale.

Bundle Adjust Residuals (Linear Scale)

Bundle Adjust Initial and Final Residuals (Linear Scale)

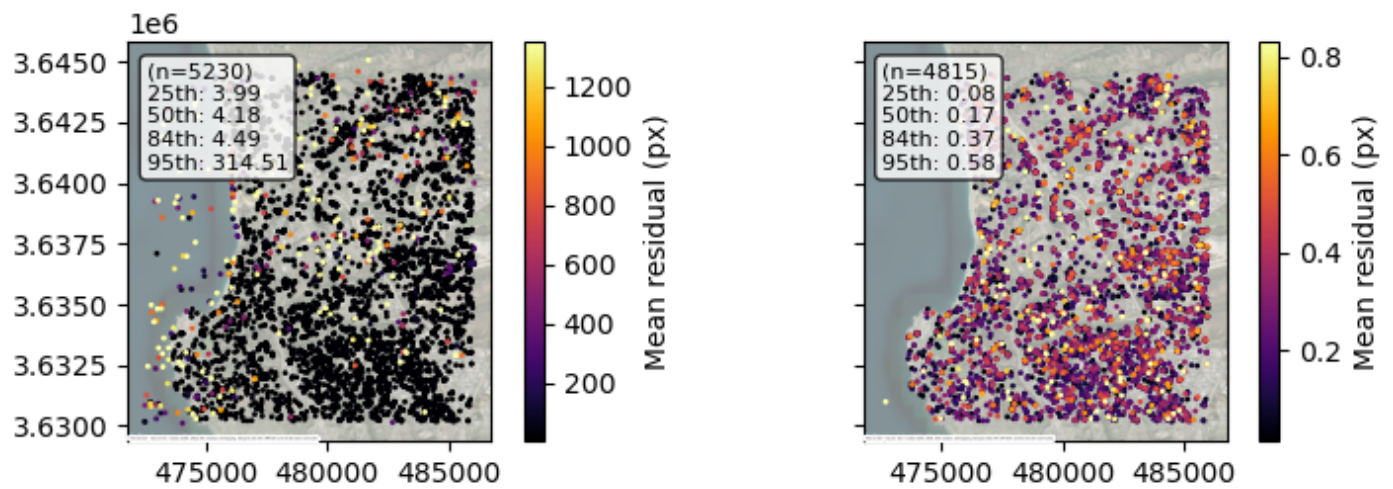


Figure 5: Initial and final bundle adjustment residuals on a linear scale.

Camera Changes from Bundle Adjustment

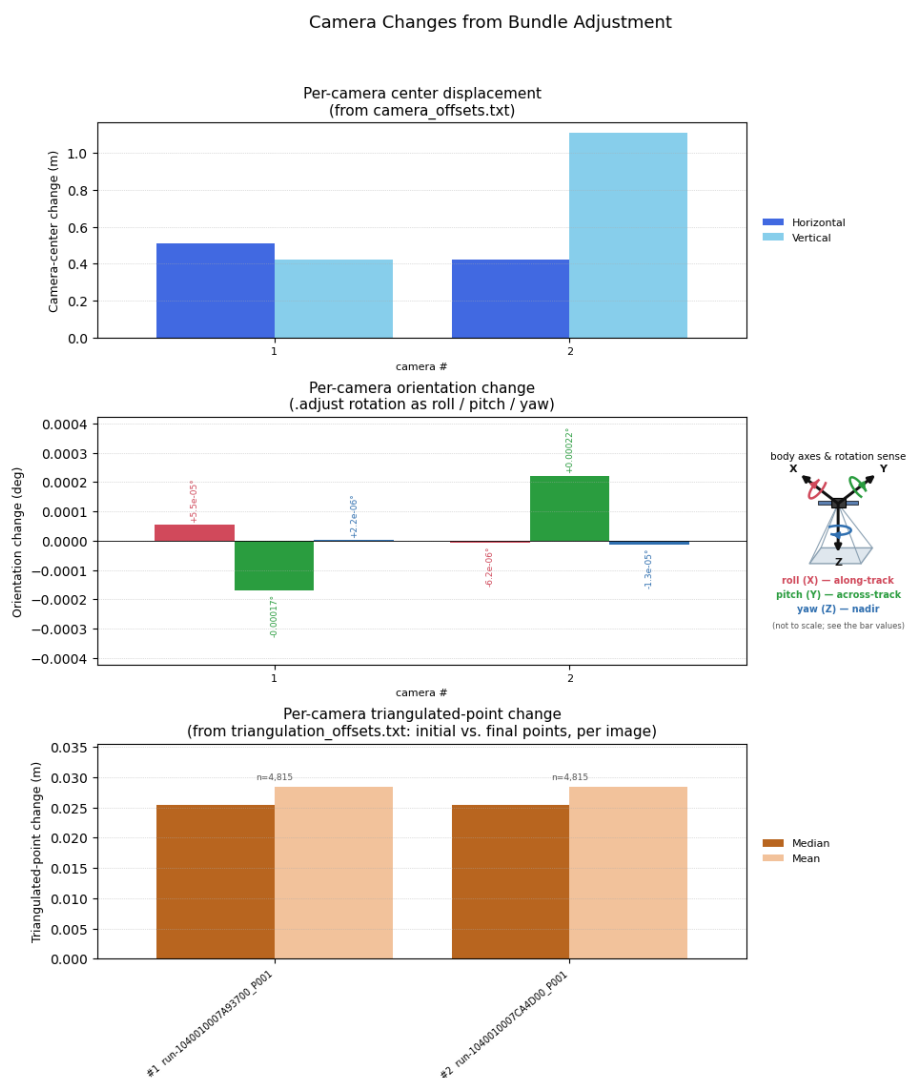


Figure 6: Per-camera changes made by `bundle_adjust`, read from the run's own outputs (the `.adjust` files and, when present, `camera_offsets.txt` and `triangulation_offsets.txt`); no original cameras are needed. Top: horizontal and vertical change of each camera center in meters (from `camera_offsets.txt` when ASP wrote it, otherwise from the `.adjust` translation). Middle: change in camera orientation in degrees as roll (about the along-track X axis), pitch (about the across-track Y axis), and yaw (about the nadir Z axis), with the value printed on every bar; the satellite cartoon is a legend for the axes and rotation sense and is not to scale. Bottom, only when `ASP >= 3.6` wrote `triangulation_offsets.txt`: median and mean change of each image's triangulated points in meters, i.e. the effect of the camera change on the ground. A large camera-center shift next to a small ground change means the solver traded camera position against orientation; the ground row is what matters for the DEM. Cameras are numbered in `.adjust` file order; the bottom row prints each camera id. A run that applied only an identity transform shows a 'no camera change' note.

Disparity

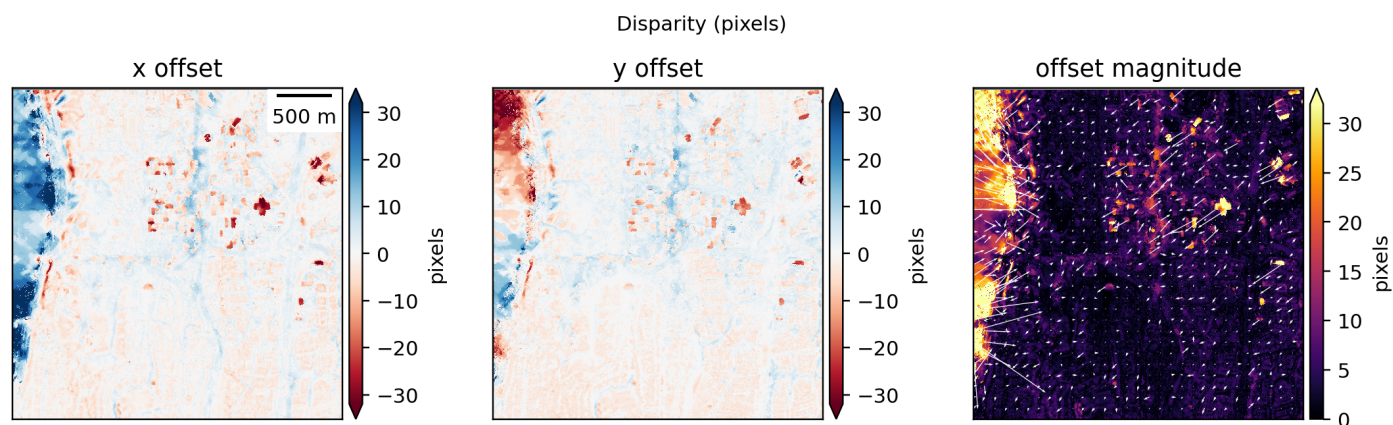


Figure 7: Horizontal and vertical disparity maps in pixels with quiver overlay.

DEM Results

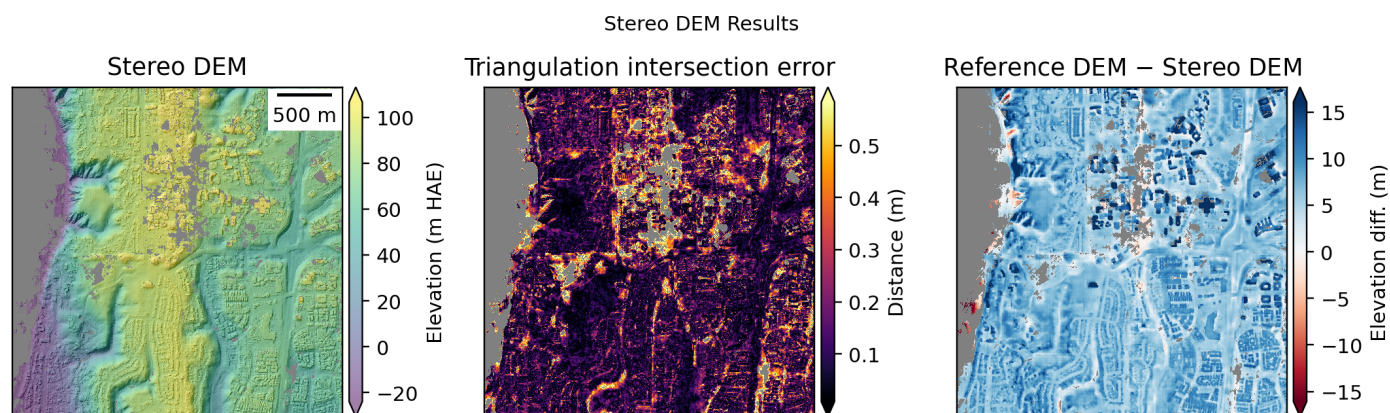


Figure 8: Output DEM with intersection error map and difference relative to the reference DEM used in processing.

Detailed Hillshade

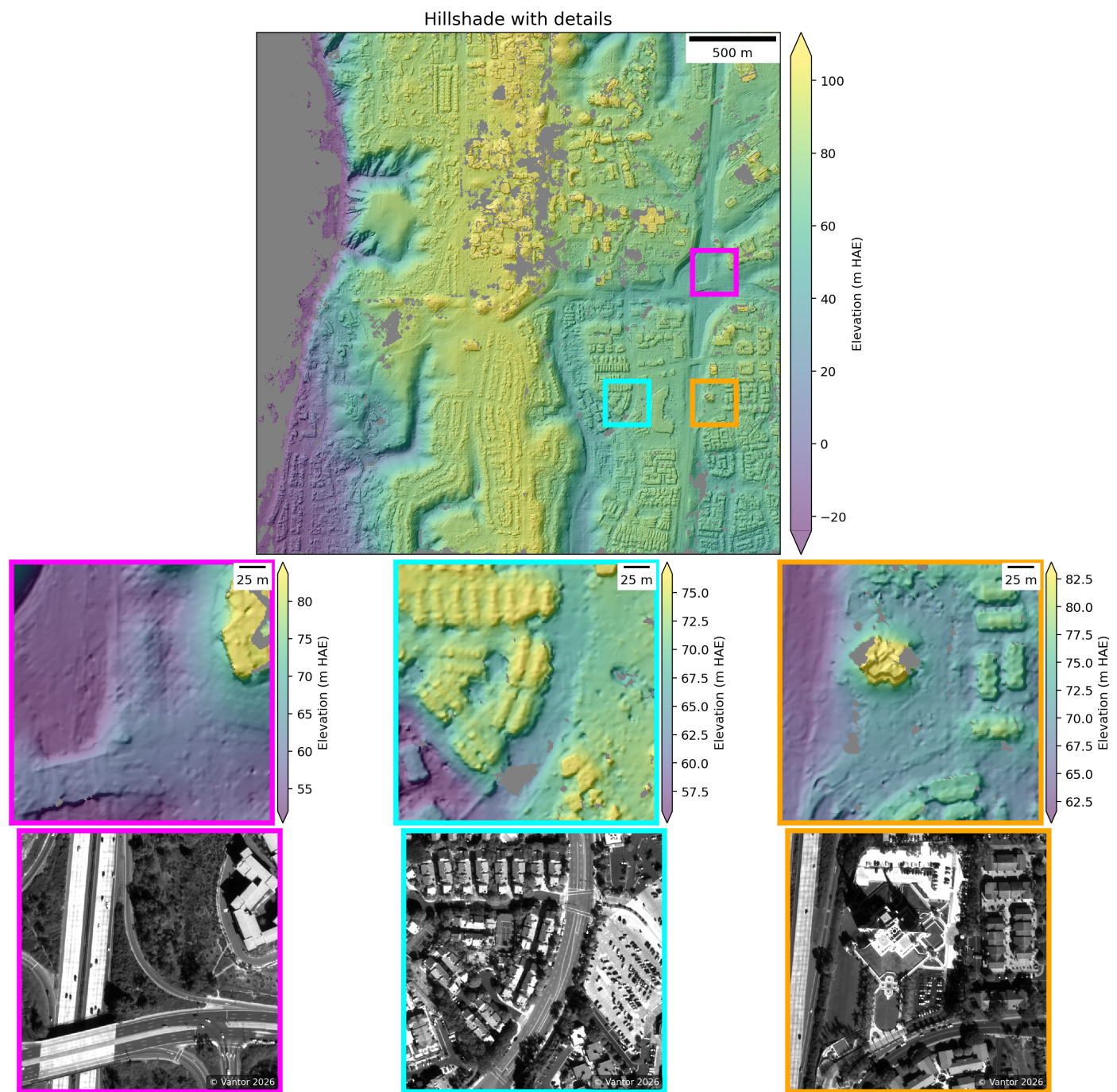


Figure 9: DEM hillshade. If the intersection error is available, zoomed subsets selected from low, medium, and high (left to right) uncertainty areas are displayed in the second row. If the mapprojected image is available, corresponding ortho image subsets are displayed in the bottom row.

ICESat-2 ATL06-SR Map

ICESat-2 ATL06-SR
all (n=1086)
2018-10-14 to 2026-07-30

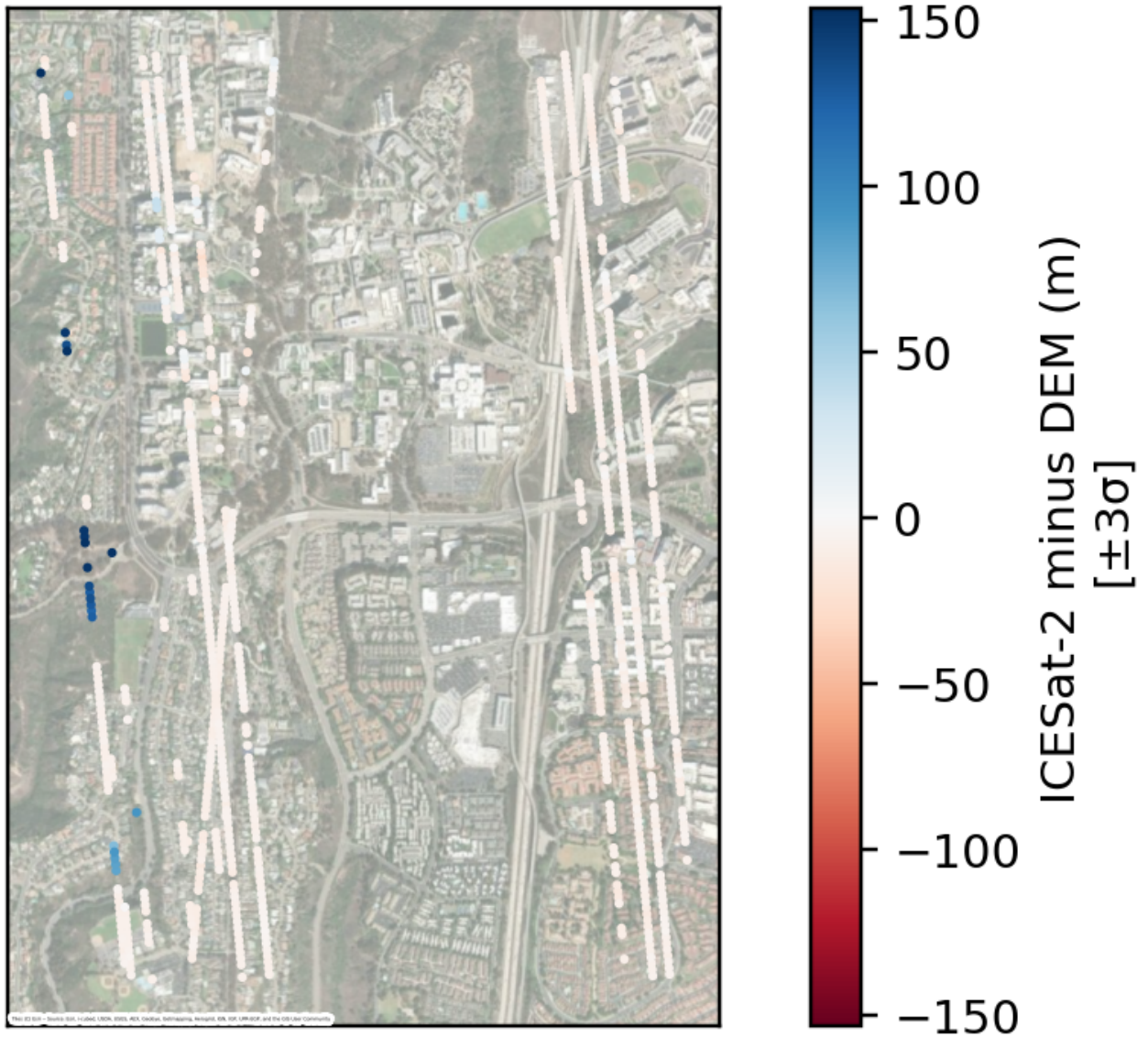


Figure 10: ICESat-2 ATL06-SR elevation differences vs. ASP DEM.

ICESat-2 ATL06-SR Histogram

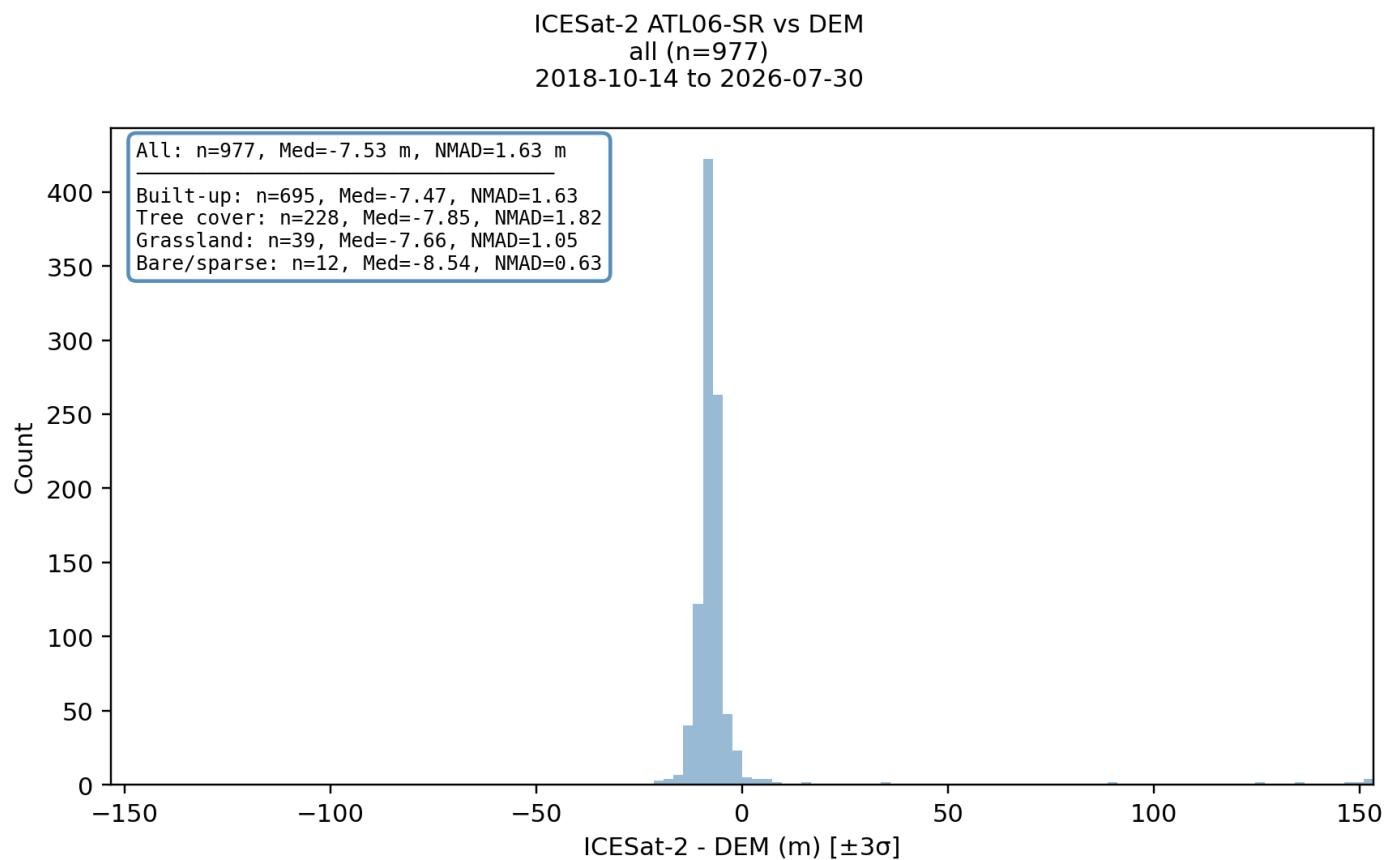


Figure 11: Distribution of elevation differences between ICESat-2 ATL06-SR and ASP DEM with per-landcover statistics.

ICESat-2 ATL06-SR Profile

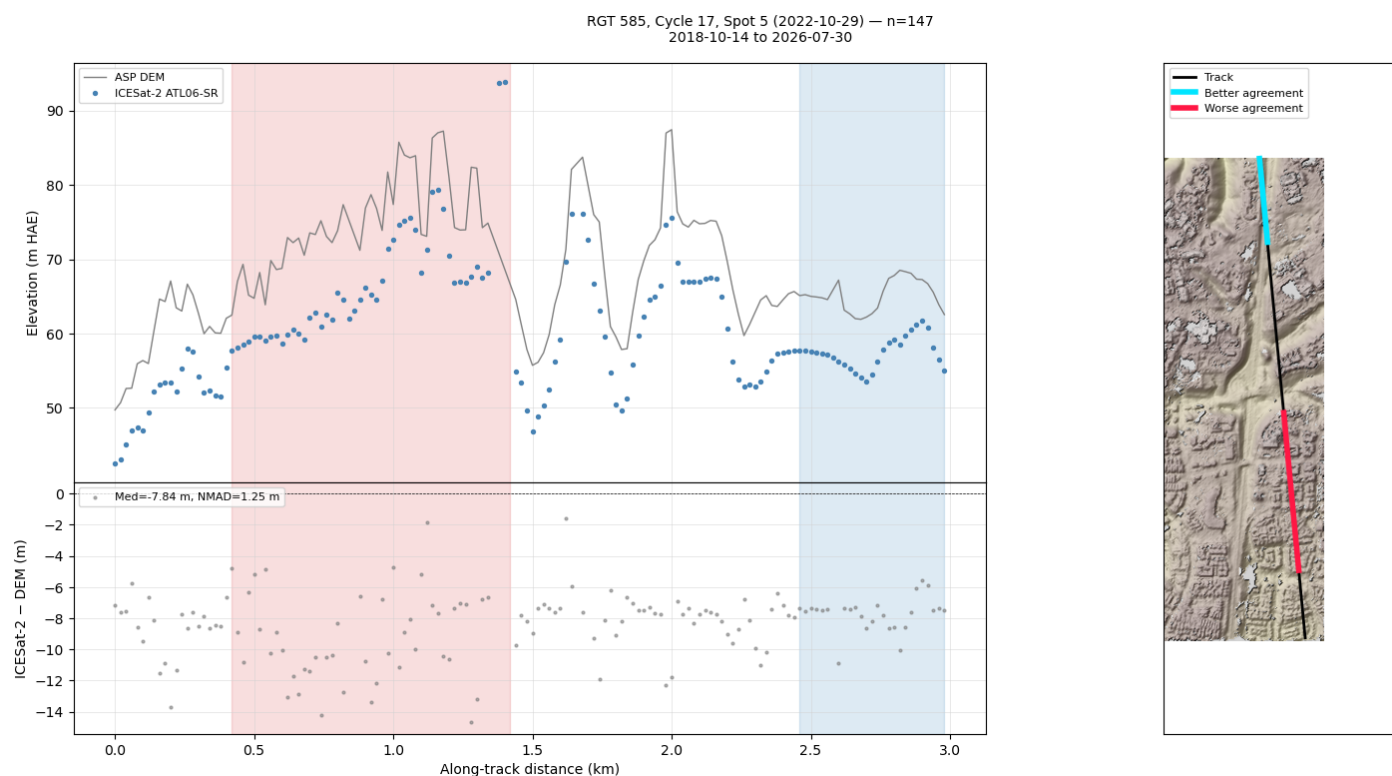


Figure 12: Elevation profile along the ICESat-2 track with the most valid points, comparing ATL06-SR and DEM heights (top) and height differences (bottom), with a context hillshade map on the right. Blue/red highlights mark the 1 km segments with better and worse agreement between ICESat-2 and the DEM.

ICESat-2 ATL06-SR Agreement Segments

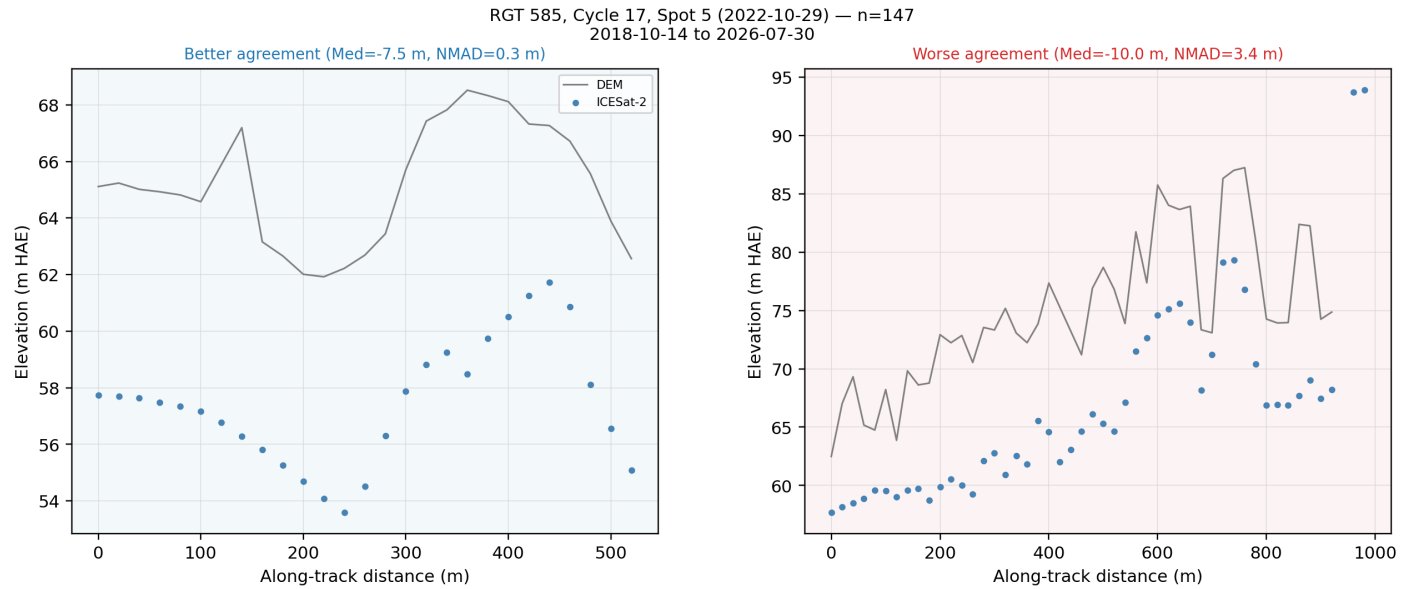


Figure 13: 1 km segments along the ICESat-2 track with better (left) and worse (right) agreement with the DEM. Segments are scored by $3 \cdot |\text{median}(dh)| + \text{NMAD}(dh)$, which weights the median bias three times more than the dispersion so that a segment with a large bias is not selected as 'better agreement' just because its NMAD is small.

DEM Alignment with ICESat-2

Alignment Parameters

Parameter	Value
processing_level	all
minimum_points	500
agreement_threshold	0.25
min_translation_threshold	0.1
improvement_threshold_pct	5.0

Error Statistics (m)

Statistic	Before	After	Change
Median	7.44	1.11	-85.0%
NMAD	1.59	1.23	-22.4%
RMSE	7.91	3.53	-55.3%

Translation (m)

Component	Value
North	0.87
East	-2.23
Down	-7.63
Magnitude T	8.00

ASP's pc_align estimates a rigid 3D translation that minimizes the height residuals between the ASP DEM and ICESat-2 ATL06-SR ground-track points used as the reference point cloud. The translation is applied to the DEM directly (geotransform + pixel-value shift, no resampling) to produce the aligned DEM.

Alignment Parameters (above):

- processing_level: ATL06-SR filter key used as the reference; 'all' uses every filtered point.
- minimum_points: minimum ATL06-SR point count required; fewer points skips the alignment.
- agreement_threshold: maximum relative disagreement across temporal sub-filters before the aligned DEM is flagged as inconsistent.
- min_translation_threshold: minimum translation magnitude (as a fraction of the DEM GSD) required to write out an aligned DEM.
- improvement_threshold_pct: minimum percentage reduction in the median residual required to keep the aligned DEM on disk; below this, the aligned DEM is removed.

Error Statistics (above, in meters): each row is one statistic of the DEM-vs-ICESat absolute height residuals, Before and After alignment; Change is the percent change from Before to After, so negative is an improvement.

- Median: median residual.
- NMAD: normalized median absolute deviation; a robust spread, insensitive to outliers.
- RMSE: root-mean-square residual; outlier-sensitive, so a large RMSE next to a small NMAD flags a tail of gross errors rather than a broad misfit.
- p16 / p50 / p84: shown instead of the above when the pc_align log predates ASP 3.7.0 (which added the Median / NMAD / RMSE summary): 16th / 50th / 84th percentile of the residuals; p50 is the median.

Translation (above, in meters): the applied translation vector as North / East / Down components and its magnitude |T|. Positive Down means the DEM was too high.

Median improved from 7.44 m -> 1.11 m (85.0% reduction). Aligned DEM written to /Users/ben/Desktop/asp-plot-examples/ucsd_stereo_21deg_12d/stereo/run-DEM_pc_align_translated.tif.

ICESat-2 ATL06-SR Histogram (Aligned DEM)

ICESat-2 ATL06-SR vs DEM
all (n=977)
2018-10-14 to 2026-07-30

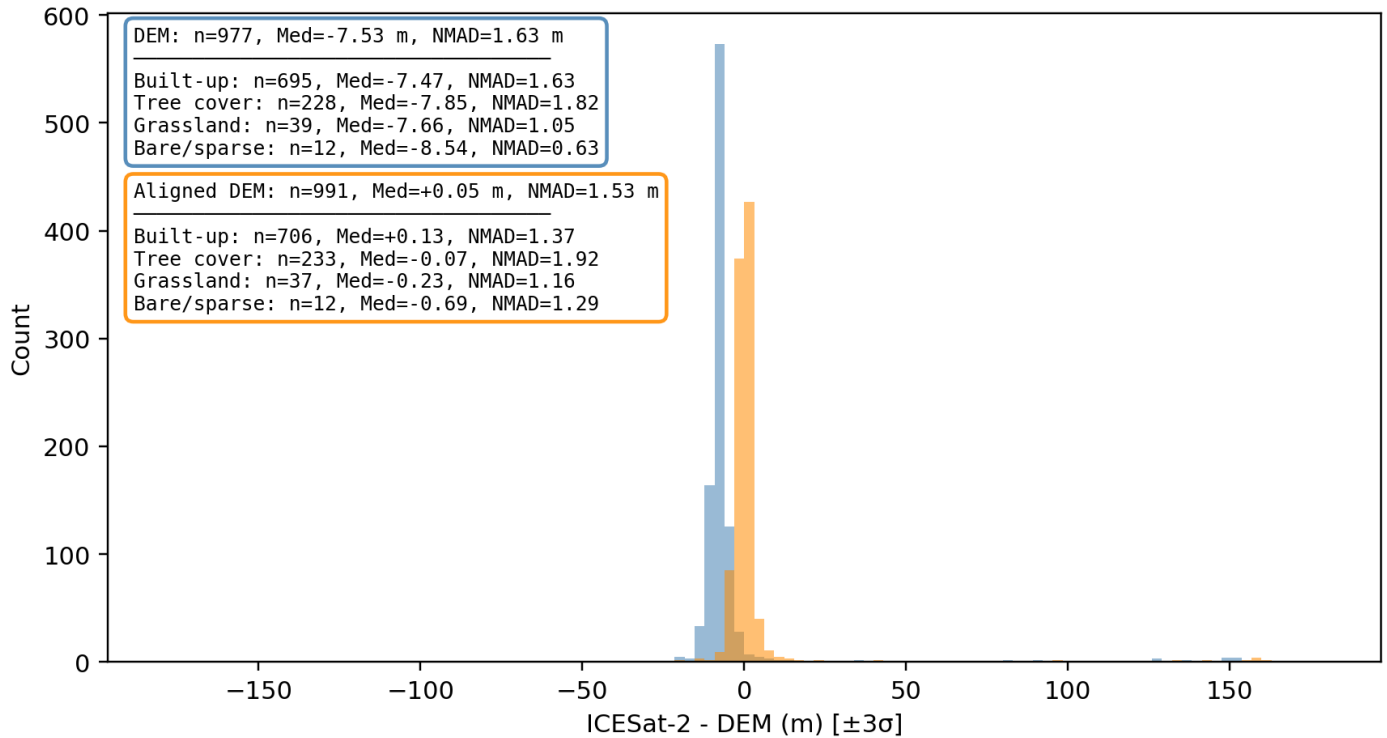


Figure 15: Pre- (steelblue) and post-alignment (orange) distributions of ICESat-2 minus DEM height differences, with per-landcover statistics in the two stacked text boxes. Box outline color matches the bar color.

ICESat-2 ATL06-SR Profile (Aligned DEM)

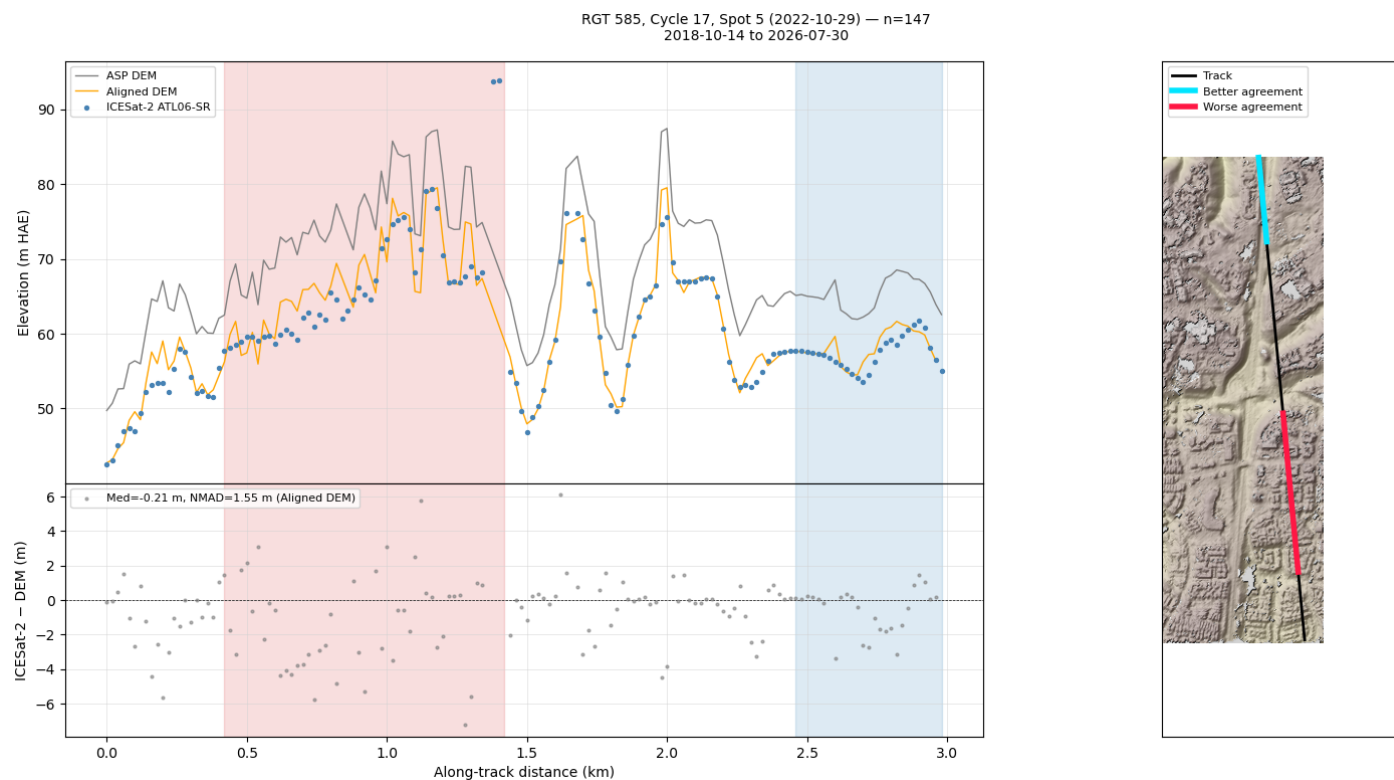


Figure 16: Elevation profile along the ICESat-2 track after *pc_align*. The aligned DEM is overlaid on the profile and used to recompute the height differences shown in the lower panel.

ICESat-2 ATL06-SR Agreement Segments (Aligned DEM)

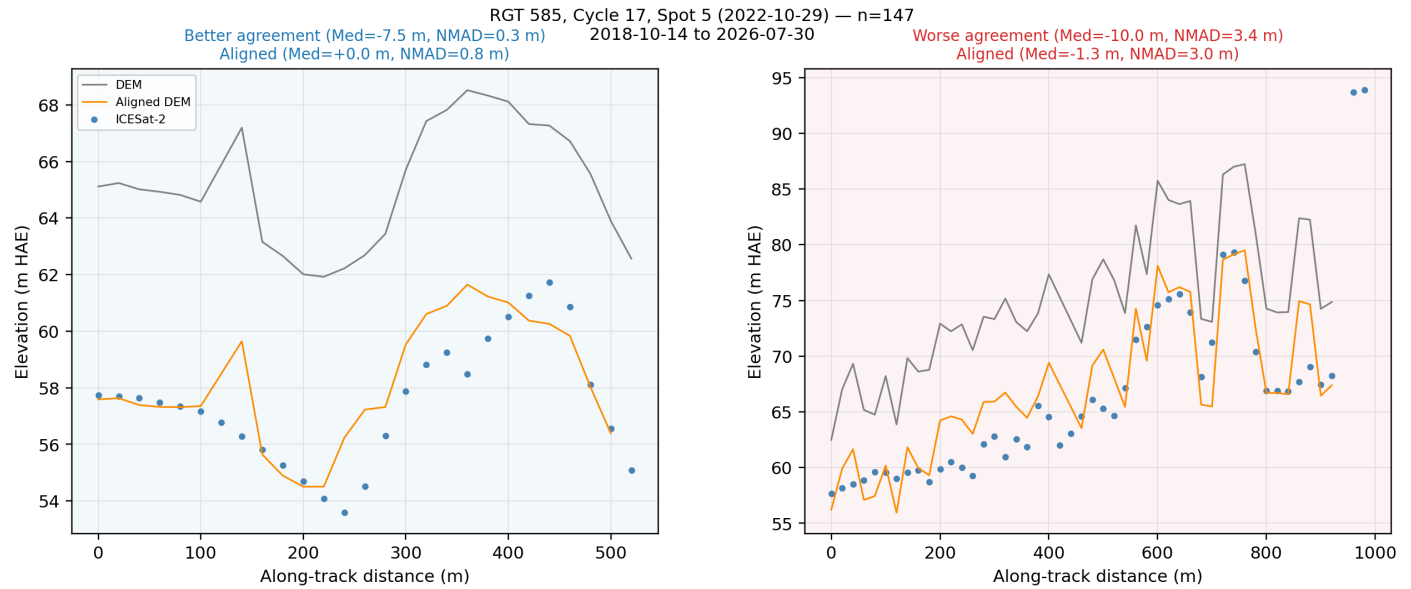


Figure 17: The same better- and worse-agreement segments as above, now with the aligned DEM overlaid. Segment selection is held fixed so Median/NMAD can be compared directly.